

CCSU BCA 1st Semester – Computer Architecture

2026 Exam-Oriented Important Notes | BCA 1002 T

The CCSU 2025–26 BCA structure lists Computer Architecture as BCA 1002 T. The syllabus covers four major units: digital principles/number systems; combinational & sequential circuits; basic computer organization/CPU design; and pipeline, I/O & memory organization.

UNIT I – Digital Principles & Number Systems

- Digital signals and digital computers: digital systems represent information using discrete values, commonly 0 and 1.
- Von Neumann architecture: input, output, memory, ALU and control unit; stored-program concept.
- Boolean algebra: basic laws such as identity, null, idempotent, complement, commutative, associative, distributive and De Morgan's laws.
- Logic gates: AND, OR, NOT, NAND, NOR, XOR and XNOR. Learn symbols, Boolean expressions and truth tables.
- K-Map: 2, 3 and 4-variable Karnaugh maps; grouping 1s/0s; SOP and POS simplification; don't-care conditions.
- Number systems: Decimal, Binary, Octal and Hexadecimal; conversions among them.
- Binary arithmetic: addition, subtraction and basic multiplication/division.
- Codes: BCD, ASCII, EBCDIC, Excess-3 and Gray code; basic error-detecting/correcting concepts.

UNIT II – Combinational & Sequential Circuits

- Half Adder: adds two 1-bit inputs; $\text{Sum} = A \oplus B$, $\text{Carry} = A \cdot B$.
- Full Adder: adds A, B and Carry-in; learn truth table and circuit.
- Half/Full Subtractor: difference and borrow.
- Encoder and Decoder: understand purpose, block diagram and truth table.
- Multiplexer (MUX): selects one of many inputs. Demultiplexer (DEMUX): routes one input to one of many outputs.
- Flip-flops: SR, D, JK and T; learn truth tables and characteristic behavior.
- Registers: storage elements made from flip-flops; shift registers and their types.
- Counters: asynchronous (ripple) and synchronous counters; basic working and differences.

UNIT III – Basic Computer Organization & CPU Design

- Instruction code and instruction format; opcode and operand concepts.
- CPU registers: general-purpose and special-purpose registers; accumulator, program counter, instruction register etc.
- Timing and control: clock, timing signals and control signals.
- Instruction cycle: fetch, decode and execute; understand the sequence.
- Memory-reference instructions and basic accumulator-based organization.
- I/O organization: input-output operations and interrupts.
- General register organization and stack organization.
- Addressing modes: immediate, direct, indirect and other common modes; learn meaning with examples.
- RISC vs CISC: instruction complexity, number of instructions, execution approach and typical design differences.
- Program control and data manipulation instructions.

UNIT IV – Pipeline, I/O & Memory Organization

- Parallel processing: performing multiple operations simultaneously to improve performance.
- Pipelining: divides instruction processing into stages; learn throughput, speed-up and pipeline hazards at a basic level.
- Arithmetic pipeline and instruction pipeline; RISC pipeline.

- Peripheral devices and I/O interface.
- DMA (Direct Memory Access): allows data transfer between I/O and memory with limited CPU involvement.
- Interrupts and priority interrupts: why they occur and how priority can be handled.
- Memory hierarchy: registers → cache → main memory → secondary storage; speed, cost and capacity trade-offs.
- Cache memory: purpose, locality and basic mapping concepts.
- Virtual memory: gives programs an abstraction of a larger memory space using secondary storage.
- Associative memory and basic memory-management hardware.

■ MOST IMPORTANT FOR EXAM

- K-Map (2/3/4 variables), SOP/POS and Boolean laws.
- Number-system conversions and binary arithmetic.
- Logic-gate truth tables and universal gates NAND/NOR.
- Half Adder and Full Adder; Encoder/Decoder; MUX/DEMUX.
- SR, D, JK and T flip-flops.
- Registers and synchronous vs asynchronous counters.
- Instruction cycle, registers, instruction formats and addressing modes.
- RISC vs CISC.
- Pipelining and basic performance concepts.
- DMA, interrupts, cache memory, memory hierarchy and virtual memory.

Important Questions – 2026 Revision

1. Explain Von Neumann architecture with a neat diagram.
2. State and explain important Boolean algebra laws and De Morgan's theorems.
3. Simplify a Boolean expression using K-Map.
4. Convert numbers between binary, decimal, octal and hexadecimal.
5. Explain ASCII, EBCDIC, BCD, Excess-3 and Gray code.
6. Explain Half Adder and Full Adder with truth tables.
7. Explain MUX, DEMUX, Encoder and Decoder.
8. Explain SR, D, JK and T flip-flops with truth tables.
9. Explain registers and synchronous/asynchronous counters.
10. Explain instruction cycle and types of CPU registers.
11. Explain addressing modes with examples.
12. Compare RISC and CISC.
13. Explain pipelining and its advantages.
14. Explain DMA and interrupt-driven I/O.
15. Explain memory hierarchy, cache memory and virtual memory.

Note: These are exam-oriented revision notes based on the CCSU 2025–26 BCA 1002 T topic structure. They are not a prediction of the exact question paper.